
DS 3001 Final Project Paper

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Abstract

Our Project explores refugee migration rates and how we can predict such rates based on origin country and asylum country freedom indexes.

1. Data

The data in our project consists of refugee migration statistics from United Nations High Commission for Refugees and Freedom Indexes from the Freedom House. With this Data we plan to investigate refugee rates from countries around the world, and eventually create a model that predicts migration rates based on the freedom indexes of ones home and host country.

The Freedom in the World dataset, published annually by Freedom House, provides scores and ratings from 2013 to 2021 for countries and territories. The data include measures such as political rights ratings, civil liberties ratings, and aggregate scores for subcategories like electoral process, political pluralism and participation, rule of law, and freedom of expression and belief. Each country-year entry also includes an overall status of Free, Partly Free, or Not Free. These variables allow us to capture the political and civil environment in both origin and asylum countries.

The UNHCR Persons of Concern dataset contains annual records beginning in 2004, organized by year, country of origin, and country of asylum. For each country pair, the dataset reports counts of various displaced groups, including refugees, asylum-seekers, internally displaced persons (IDPs), stateless persons, and others of concern. These values reflect the scale of displacement across borders and within states. These values provide the dependent variable in our analysis, which is the amount of refugee movement from one country to another in a given year.

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The unit of analysis in our study is the country-year pair, which links refugee flows from an origin country to an asylum country in a given year with their corresponding freedom index values. This allows us to test hypotheses about how political repression, democratic governance, and civil liberties shape refugee migration. While the UNHCR data captures migration outcomes, the Freedom House data provides measures of the political conditions that may act as push and pull factors.

These datasets do consist limitations, such that refugee statistics often underrepresented the quantities of undocumented displacement, and the Freedom House scores, while widely used, are based partly on subjective assessments. Despite this, their examination provides a unique opportunity to connect quantitative migration patterns with qualitative assessments of political freedom.

1.1. Data Cleaning

The data cleaning process for the refugee and freedom index data involved several steps to ensure the data was ready for analysis. For the persons-of-concern.csv file, which contains the data regarding refugee migration, the initial loading and inspection revealed no immediate issues with missing values or data types, so no further cleaning was required. For the All-data-FIW-2013-2021.xlsx file, which contains the data regarding freedom indexes, initial attempts to load the data resulted in incorrect headers due to metadata at the beginning of the sheet. This was corrected by specifying the correct sheet name ('FIW13-21') and header row (row 0) during loading. The columns were then renamed for clarity, specifically 'C/T?' to 'Country Type', 'Country/Territory' to 'Country', and 'Total' to 'Freedom Total Score'. Finally, the first row, which contained descriptive information rather than data. We removed this so data handling could be smoother. Then relevant columns were selected and converted to appropriate numeric types. Finally we checked for null values and confirmed that there were no missing values in the selected columns after these steps.

1.2. Data Dictionary

This table provides a data dictionary for the Freedom in the World (FIW) dataset, defining the various columns and their meanings.

Table 1. Data Dictionary

Column Name	Description
Year	The year of the data record.
Country of Asylum	The name of the country where a person is seeking or has received protection.
Country of Origin	The name of the country from which a person has been displaced.
Country of Asylum ISO	The ISO 3166-1 alpha-3 code for the country of asylum.
Country of Origin ISO	The ISO 3166-1 alpha-3 code for the country of origin.
Refugees	The total number of refugees.
Asylum-seekers	The total number of asylum-seekers.
IDPs	The total number of Internally Displaced Persons.
Other people in need of international protection	The total number of people in need of international protection who do not fall into the other categories.
Stateless persons	The total number of stateless persons.
Host community	The total number of individuals in the host community.
Others of concern	The total number of individuals who are not included in the main categories but are of concern to humanitarian organizations.
Country/Territory	The name of the country or territory.
Region	The geographic region of the country or territory.
C/T	Indicates if the entry is a country (c) or territory (t).
Edition	The year of the report.
Status	The overall freedom status: F (Free), PF (Partly Free), or NF (Not Free).
PR rating	The Political Rights Rating, on a scale of 1 (least free) to 7 (most free).
CL rating	The Civil Liberties Rating, on a scale of 1 (least free) to 7 (most free).
PR	The aggregate score for the Political Rights category, based on sub-scores from A, B, and C.
A1 to A3	Sub-scores for the A. Electoral Process subcategory.
A	The aggregate score for the A. Electoral Process subcategory.
B1 to B4	Sub-scores for the B. Political Pluralism and Participation subcategory.
B	The aggregate score for the B. Political Pluralism and Participation subcategory.
C1 to C3	Sub-scores for the C. Functioning of Government subcategory.
C	The aggregate score for the C. Functioning of Government subcategory.
CL	The aggregate score for the Civil Liberties category, based on sub-scores from D, E, F, and G.
D1 to D4	Sub-scores for the D. Freedom of Expression and Belief subcategory.

2. Methods and Results

2.1. Method Overview

The primary objective of our project is to create a predictive model that quantifies the relationship between political and civil liberties (push and pull factors) with refugee migration flows. We are framing our task as a **regression problem**, where the dependent variable is the count of refugees from a given origin country to an asylum country in a specific year. The core methodology involves three phases: feature engineering and data integration, predictive model training, and model validation and assessment. Due to the inherent non-linear and potentially discontinuous nature of political conditions and their impact on human migration, we will employ a non-parametric approach. This means that we will use the aggregated total freedom scores of the origin and asylum countries as the key independent variables (features) to predict the total number of refugees between them.

FEATURE ENGINEERING AND DATA INTEGRATION

To align the two datasets, the time-series refugee counts from UNHCR and the annual freedom index scores from Freedom House, we performed a key data integration step. For every unique country-year observation in the UNHCR data (defined by Year, Country of Origin ISO, and Country of Asylum ISO), we joined the corresponding freedom scores from the Freedom House data. This join was executed twice:

1. To associate the **Origin Country's** Total freedom score (F_{Origin}) based on the Year and Country of Origin. The Origin Country's score is hypothesized to act as a **push factor** (low score predicts high migration).
2. To associate the **Asylum Country's** Total freedom score (F_{Asylum}) based on Year and Country of Asylum. The Asylum Country's score is hypothesized to act as a **pull factor** (high score predicts high immigration).

The target variable for this model, **Refugees**, is a count, and as expected, highly skewed, with many zero or near-zero values. To create a more normal and stable variance for the regression model, we applied a logarithmic transformation: $Y = \log(1 + \text{Refugees})$. The addition of 1 ensures that zero refugee counts can be handled without resulting in undefined logarithms. The final engineered dataset consists of the features (F_{Origin} , F_{Asylum}) and the target variable Y . The merged dataset includes approximately 99,946 unique country-pair-year observations spanning 2013–2021.

2.2. Model Justification

We selected a **Random Forest Regressor (RFR)** as our predictive model. [Insert Image of our Decision Tree Structure]

This choice is justified by a few advantages over traditional linear models, especially for our data:

1. **Non-linearity and Interactions:** The relationship between political freedom and migration is complex; therefore, it is unlikely to be a simple linear function. The RFR, being based on decision trees, can inherently capture non-linear relationships and intricate interaction effects between the origin and asylum country scores without requiring manual feature engineering.
2. **Robustness to Data Characteristics:** RFR is robust against outliers and requires minimal data preprocessing. Specifically the process avoids the need for feature scaling, which is beneficial when combining features with different inherent scales, such as the aggregated freedom scores that come from very different countries/circumstances.
3. **Feature Importance:** The RFR also provides a quantitative measure of **feature importance**, allowing us to determine whether the push factors (F_{Origin}) or the pull factors (F_{Asylum}) have a greater predictive influence on the migration rate.

2.3. Model Training Procedure

DATA SPLITTING

The engineered dataset was randomly partitioned into training and testing sets using an 80/20 split. The training set (80%) was used to train the RFR model, and the held-out test set (20%) was reserved exclusively for final evaluation of the model’s performance on unseen data. While a temporal split is possible for time-series data, a random split was chosen to maximize the diversity of country-year observations in both the training and testing phases. Since some of the countries had conflicts at different years, a temporal split may also have yielded results that were ignorant of the conflict periods.

HYPERPARAMETER OPTIMIZATION

To prevent overfitting and optimize the model’s generalization, we performed **cross-validated hyperparameter tuning**. We used a 5-fold cross-validation and a Randomized Search to efficiently explore the hyperparameters. Key hyperparameters targeted for optimization included:

- **n_estimators:** The number of decision trees in the forest.
- **max_depth:** The maximum depth of each individual decision tree.
- **min_samples_split:** The minimum number of samples required to split an internal node.

During cross-validation, the model’s performance was evaluated using the negative Mean Squared Error (MSE) on the log-transformed target, with the goal of selecting the parameter combination that minimizes MSE.

2.4. Model Validation Plan

The trained Random Forest Regressor will be evaluated on the withheld test set using both statistical and domain-specific metrics.

STATISTICAL METRICS (LOG SCALE)

The primary statistical metrics calculated on the log-transformed target variable (Y) and its prediction ($\hat{Y}$) are R^2 and **Root Mean Squared Error (RMSE)**.

DOMAIN-SPECIFIC METRICS (ORIGINAL SCALE)

To ensure the model’s predictions are interpretable in the context of real-world refugee counts, the predictions $\hat{Y}$ will be inverse-transformed back to the original scale using the formula:

$$\text{Predicted Refugees} = \exp(\hat{Y}) - 1$$

The model’s final performance will be assessed using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE).

3. Conclusion

Impact Statement

References

- Freedom House. Freedom in the world. Technical report, Freedom House, Washington, DC, 2025. URL <https://freedomhouse.org/report/freedom-world>.
- UNHCR. Refugee statistics – operational data portal. Technical report, United Nations High Commissioner for Refugees, Geneva, Switzerland, 2025. URL <https://www.unhcr.org>.
- World Bank. World development indicators. Technical report, The World Bank, Washington, DC, 2025. URL <https://databank.worldbank.org/source/world-development-indicators>.

A. Appendix

The `\onecolumn` command above can be kept in place if you prefer a one-column appendix, or can be removed if you prefer a two-column appendix. Apart from this possible change, the style (font size, spacing, margins, page numbering, etc.) should be kept the same as the main body.